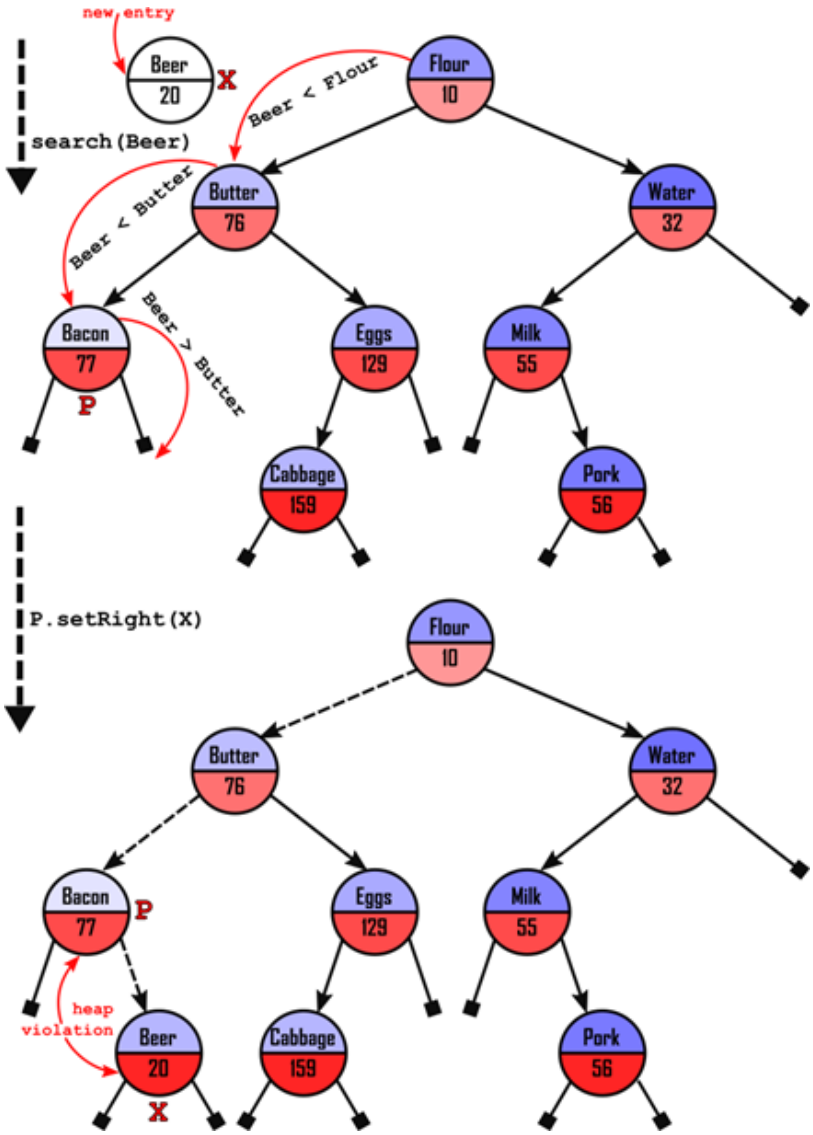


Figure 3.8 First steps of inserting a new entry. We search the new entry's key to spot the right place to add a new leaf in the tree. Now we must check if the new node violates the heap's invariants: unfortunately, in this example it does, so we need to correct the situation by performing a right rotation.

Listing 3.5 method insert

```
function insert(treap, key, priority)
                  #1
```